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Portfolio Problems for Test 3

Exercise 6.2.10

Let $[a,b] \subset \mathbb{R}$ be a closed interval, and let $\{f_n\}$ be a sequence of functions on $[a,b]$ such that $f_n(x) \to f(x)$ pointwise on $[a,b]$, each f_n is increasing on $[a,b]$, and the limit function f is continuous on $[a,b]$. Prove that $f_n \to f$ uniformly on $[a,b]$.

Proof.

Let $\varepsilon > 0$ be given.

Because f is continuous on a closed interval, it is uniformly continuous. So we can pick $\delta > 0$ such that

$$|x - x'| < \delta \;\Rightarrow\; |f(x) - f(x')| < \frac{\varepsilon}{3}.$$

Choose a partition $a = x_0 < x_1 < x_2 < \cdots < x_m = b$ so that each subinterval has length $\leq \delta$.

At each point x_i the sequence converges: $f_n(x_i) \to f(x_i)$. Hence we can find N_i so that for all $n \geq N_i$,

$$|f_n(x_i) - f(x_i)| < \frac{\varepsilon}{3}.$$

Let $N = \max\{N_0, N_1, \ldots, N_m\}$.

Take any $x \in [a,b]$. Then x lies in some subinterval $[x_{i-1}, x_i]$.

Since each f_n is increasing, $f_n(x_{i-1}) \leq f_n(x) \leq f_n(x_i)$. Thus,

$$|f_n(x) - f(x_i)| \leq |f_n(x_{i-1}) - f_n(x_i)|.$$

Then we can bound $|f_n(x) - f(x)|$ using the triangle inequality:

$$\begin{aligned} |f_n(x) - f(x)| &\leq |f_n(x) - f_n(x_i)| + |f_n(x_i) - f(x_i)| + |f(x_i) - f(x)| \\ &\leq |f_n(x_{i-1}) - f_n(x_i)| + \frac{\varepsilon}{3} + \frac{\varepsilon}{3}. \end{aligned}$$

By the uniform continuity of f, since $|x_i - x_{i-1}| \leq \delta$, we have $|f(x_i) - f(x_{i-1})| < \frac{\varepsilon}{3}$. Moreover, for $n \geq N$,

$$|f_n(x_i) - f(x_i)| < \frac{\varepsilon}{3} \quad \text{and} \quad |f_n(x_{i-1}) - f(x_{i-1})| < \frac{\varepsilon}{3}.$$

Hence,

$$\begin{aligned} |f_n(x_{i-1}) - f_n(x_i)| &\leq |f_n(x_{i-1}) - f(x_{i-1})| + |f(x_{i-1}) - f(x_i)| + |f(x_i) - f_n(x_i)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Therefore,

$$|f_n(x) - f(x)| < \varepsilon + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \frac{5\varepsilon}{3}.$$

Since $\varepsilon > 0$ was arbitrary, we can replace ε with $\frac{3\varepsilon}{5}$ in the initial choices of δ and N. Doing so yields $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$ and all $x \in [a,b]$.

$\therefore$ The convergence is uniform on $[a,b]$. $\square$

Exercise 6.5.4a

Assume $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges on $(-R, R)$; show that $F(x) = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ is defined on $(-R, R)$ and satisfies $F'(x) = f(x)$.

Proof.

Suppose $f(x) = \sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $R > 0$. Define

$$F(x) = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}, \qquad x \in (-R, R).$$

We show that F is well-defined on $(-R, R)$ and that $F'(x) = f(x)$ for every $x \in (-R, R)$.

Let $x \in (-R, R)$ be arbitrary. Choose r such that $|x| < r < R$. Since the original series converges at $x = r$ (because $|r| < R$), Theorem 6.5.1 implies that the series converges absolutely for all x with $|x| < r$. In particular, $\sum_{n=0}^{\infty} |a_n| \, |x|^n$ converges.

For the series defining F, we have:

$$\sum_{n=0}^{\infty} \left| \frac{a_n}{n+1} x^{n+1} \right| = |x| \sum_{n=0}^{\infty} |a_n| \, |x|^n \cdot \frac{1}{n+1} \leq |x| \sum_{n=0}^{\infty} |a_n| \, |x|^n.$$

Since the right-hand side converges, the comparison test yields the absolute convergence of the series for F at x. Hence $F(x)$ is defined for every $x \in (-R, R)$.

To differentiate F, write F as a standard power series:

$$F(x) = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} = \sum_{m=1}^{\infty} \frac{a_{m-1}}{m} x^m.$$

The radius of convergence of this series is also R, as the coefficients satisfy:

$$\limsup_{m \to \infty} \left| \frac{a_{m-1}}{m} \right|^{1/m} = \frac{1}{R}.$$

By the theorem on termwise differentiation of power series, for every $x \in (-R, R)$,

$$F'(x) = \sum_{m=1}^{\infty} a_{m-1} x^{m-1} = \sum_{n=0}^{\infty} a_n x^n = f(x).$$

$\therefore$ F is differentiable on $(-R, R)$ and $F' = f$.

□

Exercise 7.2.5

Assume that, for each n, f_n is integrable on $[a,b]$. If $f_n \to f$ uniformly on $[a,b]$, prove that f is also integrable on $[a,b]$.

Proof.

Assume that, for each n, f_n is an integrable function on $[a,b]$. If (f_n) converges uniformly to a function f on $[a,b]$, we prove that f is also integrable on $[a,b]$.

We first show that f is bounded on $[a,b]$. Because each f_n is integrable, it is bounded. Set $M_n = \sup_{x\in[a,b]} |f_n(x)|$. Uniform convergence implies there exists N such that for all $x \in [a,b]$, $|f_N(x) - f(x)| < 1$. Hence for every x,

$$|f(x)| \le |f_N(x)| + 1 \le M_N + 1,$$

so f is bounded. Consequently, for any partition P, the suprema and infima of f on subintervals are finite real numbers.

Now let $\varepsilon > 0$ be arbitrary. Choose $\delta = \frac{\varepsilon}{3(b-a)}$. By uniform convergence, there exists K such that for all $n \ge K$ and all $x \in [a,b]$,

$$|f_n(x) - f(x)| < \delta.$$

Fix an index $n \ge K$. Since f_n is integrable, there exists a partition $P = \{x_0, x_1, \dots, x_m\}$ of $[a,b]$ such that

$$U(f_n, P) - L(f_n, P) < \frac{\varepsilon}{3}.$$

For each subinterval $[x_{k-1}, x_k]$, define

$$M_k = \sup_{x\in[x_{k-1},x_k]} f(x), \quad m_k = \inf_{x\in[x_{k-1},x_k]} f(x),$$

and similarly $M_k^{(n)}, m_k^{(n)}$ for f_n. From the uniform bound $|f_n(x) - f(x)| < \delta$ we obtain

$$f(x) \le f_n(x) + \delta \quad \text{and} \quad f_n(x) - \delta \le f(x)$$

for all x in the subinterval. Taking suprema and infima yields

$$M_k \le M_k^{(n)} + \delta, \qquad m_k \ge m_k^{(n)} - \delta.$$

Therefore

$$M_k - m_k \le \big(M_k^{(n)} - m_k^{(n)}\big) + 2\delta.$$

Multiplying by $\Delta x_k = x_k - x_{k-1}$ and summing over $k = 1, \dots, m$ gives

$$U(f,P) - L(f,P) = \sum_{k=1}^{m} (M_k - m_k)\Delta x_k \le \sum_{k=1}^{m} \big(M_k^{(n)} - m_k^{(n)}\big)\Delta x_k + 2\delta \sum_{k=1}^{m} \Delta x_k.$$

But $\sum_{k=1}^{m} \Delta x_k = b - a$, and the first sum is precisely $U(f_n, P) - L(f_n, P)$. Hence

$$U(f,P) - L(f,P) \le \big(U(f_n,P) - L(f_n,P)\big) + 2\delta(b-a).$$

Substituting the bounds,

$$U(f,P) - L(f,P) < \frac{\varepsilon}{3} + 2 \cdot \frac{\varepsilon}{3(b-a)} \cdot (b-a) = \frac{\varepsilon}{3} + \frac{2\varepsilon}{3} = \varepsilon.$$

Thus for every $\varepsilon > 0$ there exists a partition P with $U(f,P) - L(f,P) < \varepsilon$. By the Riemann integrability criterion (Theorem 7.2.8), f is integrable on $[a,b]$.

□